SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE CODE: ITA 0605 COURSE NAME: MACHINE LEARNING

COURSE OUTCOME COVERED

CO4: K- Nearest Neighbour Learning – Locally weighted Regression – Radial Bases Functions – Case Based Learning (BL 5)

Assessment Tool Weightage: 10%

QUESTIONS: 15 Total Marks:20

Annexure A

Comparative Analysis

Code of Conduct:

I **Mathiprakash E (192312085)** certify that this submission is my original workand that I have adhered to the guidelines specified for this assessment.I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

QUESTIONS

1, Compare K-Nearest Neighbour and Case-Based Learning using a real-world example such as medical diagnosis. Analyze differences in memory usage, similarity measures, and prediction strategy.

2. Compare Locally Weighted Regression and Radial Basis Function networks for function approximation tasks. Use an example dataset to explain differences in locality, computational cost, and smoothness of predictions.

3. Analyze the differences between K-Nearest Neighbour and Radial Basis Function models in classification problems. Provide an example and compare decision boundaries, training time, and generalization ability.

4. Compare Case-Based Learning and Locally Weighted Regression using a practical example such as recommendation systems. Discuss how each approach handles new data, adaptability, and computational efficiency.

5. Compare Decision Tree Learning and K-Nearest Neighbour (KNN) using a classification problem such as customer churn prediction. Analyze differences in model interpretability, training time, handling of noisy data, and decision-making process. Discuss how each method performs with large datasets and complex feature spaces.

ANSWERS

1. K-Nearest Neighbour vs Case-Based Learning in Medical Diagnosis

K-Nearest Neighbour (KNN) and Case-Based Learning (CBL) are both instance-based learning approaches. They make predictions by using previously stored examples rather than building a completely generalized model. A suitable real-world example is a medical diagnosis system that predicts a disease based on symptoms, test results, age, blood pressure, glucose level, and other patient information.

K-Nearest Neighbour -

KNN stores patient records and compares a new patient's health information with existing patients. It calculates the similarity or distance between records and selects the K most similar patients. The diagnosis is then usually determined through majority voting. For example, if 4 out of the 5 most similar patients were diagnosed with diabetes, the new patient may also be classified as diabetic.

Case-Based Learning -

Case-Based Learning stores previous medical cases along with their diagnoses and other relevant information. When a new patient is received, the system retrieves similar previous cases. The previous diagnosis can then be reused or adapted according to the differences between the old and new patient.

Factor	**KNN**	**Case-Based Learning**
Memory usage	Stores training examples	Stores previous cases and their solutions
Similarity measure	Usually, mathematical distance	Can use domain-specific similarity
Prediction strategy	Majority voting or weighted voting	Retrieves and adapts previous cases
Adaptability	Moderate	High
Medical example	Classifies a patient based on nearby patients	Retrieves similar diagnosed patients

Analysis -

KNN is simpler because its prediction process mainly depends on distance and voting. CBL can be more sophisticated because it can consider the context of previous cases and adapt their solutions. However, both methods can require significant memory because historical cases need to be retained.

Conclusion: KNN is suitable when patient attributes can be represented numerically and classified using distance. Case-Based Learning is more suitable when previous medical cases contain detailed information that can be directly reused or adapted for new patients.

2. Locally Weighted Regression vs Radial Basis Function Networks

Locally Weighted Regression (LWR) and Radial Basis Function (RBF) networks are techniques used for function approximation. Consider a dataset containing advertising expenditure and corresponding product sales. The relationship may not be a simple straight line, so a non-linear approximation method can be useful.

Locally Weighted Regression -

LWR predicts the output by giving greater importance to training points that are close to the input being considered. Distant observations receive smaller weights. Therefore, the model focuses mainly on the local region around the query point.

For example, when predicting sales for a particular advertising expenditure, LWR gives more importance to observations with similar advertising expenditure.

Radial Basis Function Network -

An RBF network uses radial basis functions to measure the influence of different centres. Each centre has a local region of influence, and nearby inputs produce stronger responses. The outputs of these functions are combined to produce the final prediction.

Factor	Locally Weighted Regression	RBF Network
Locality	Directly focuses on nearby observations	Uses local influence around learned centres
Training	Requires little explicit training	Requires centre and weight training
Prediction cost	Can be high because data points are examined	Generally faster after training
Smoothness	Depends on bandwidth and weighting	Usually produces smooth predictions
Adaptability	Easily adapts to new observations	May require retraining for major changes
Storage	May require many training observations	Mainly uses learned centres and weights

Analysis -

LWR is highly flexible because its prediction can change according to the local data surrounding each new point. However, this flexibility can make prediction computationally expensive for large datasets. RBF networks perform more initial training but can provide efficient predictions after the network has learned its centres and weights.

Conclusion: LWR is useful when local adaptation is important, while RBF networks are better when a smooth approximation and faster repeated predictions are required.

3. K-Nearest Neighbour vs Radial Basis Function Models in Classification

KNN and Radial Basis Function (RBF) models can both solve classification problems, but they learn and make decisions differently. Consider an email classification system that classifies emails as spam or non-spam based on features such as word frequencies, message length, and other numerical characteristics.

KNN Classification -

KNN does not create a traditional classification model during training. Instead, it stores the training examples. When a new email is received, its distance from existing emails is calculated. The classes of the nearest K emails are examined, and majority voting determines the final class.

RBF Classification -

An RBF network learns representative centres from the training data. Each centre creates a radial region of influence. A new email activates the RBF units according to its similarity to these centres. The resulting values are then used to determine whether the email belongs to the spam or non-spam class.

Factor	**KNN**	**RBF Network**
Decision boundary	Can be irregular and highly data-dependent	Generally smoother
Training time	Very low	Higher
Prediction time	Can be high for large datasets	Usually faster after training
Data storage	Stores training examples	Stores learned centres and weights
Generalization	Depends strongly on K and data quality	Can generalize smoothly after training
Noise sensitivity	Sensitive to noisy neighbours	Can reduce some local noise through learned representation

Analysis -

KNN can create highly flexible decision boundaries because every new prediction depends on nearby training examples. However, this can make it sensitive to noise and irrelevant features. RBF networks create smoother boundaries because they learn a set of representative regions.

Conclusion: KNN is attractive when training must be simple and the dataset is relatively small. RBF networks are more suitable when smooth decision boundaries, faster prediction, and better generalization are required after training.

4. Case-Based Learning vs Locally Weighted Regression in Recommendation Systems

Case-Based Learning (CBL) and Locally Weighted Regression (LWR) can both be applied to recommendation systems, but they solve the recommendation problem in different ways. Consider an online shopping system that recommends products based on customer behaviour.

Case-Based Learning -

CBL stores previous customer cases, including customer preferences, products purchased, ratings, and previous recommendations. When a new customer interacts with the system, it searches for customers with similar characteristics. The recommendations associated with those cases can then be reused or adapted.

For example, if a new customer has preferences similar to several previous customers who purchased a particular laptop, the system can recommend that laptop.

Locally Weighted Regression -

LWR can be used when the system needs to predict a numerical value, such as the rating a customer may give to a product. It gives greater importance to customers whose characteristics are close to the current customer.

For example, customers with similar purchasing behaviour may receive higher weights when predicting the expected rating of a new product.

Factor	**Case-Based Learning**	**Locally Weighted Regression**
New data	Retrieves similar previous cases	Uses nearby data for prediction
Adaptability	Very high	High
Prediction	Reuses or adapts previous solutions	Produces numerical predictions
Computational efficiency	Can become expensive with many cases	Can be expensive during prediction
Main strength	Reusing previous experiences	Local numerical prediction
Suitable application	Product recommendation	Rating or demand prediction

Analysis -

CBL is useful when previous recommendations can directly provide useful solutions. It can easily adapt when new cases are added because new experiences can simply become part of the case library. LWR is more appropriate when the system needs to estimate continuous values such as ratings, demand, or expected customer spending.

Conclusion: CBL provides flexible experience-based recommendations, while LWR provides locally adapted numerical predictions. CBL is more intuitive for experience-based recommendation, whereas LWR is useful when numerical prediction is the primary requirement.

5. Decision Tree Learning vs KNN for Customer Churn Prediction

Decision Tree Learning and K-Nearest Neighbour (KNN) can both be used to predict customer churn. Consider a telecom company that wants to identify whether a customer is likely to leave the service. Features may include contract type, monthly charges, service usage, number of complaints, customer age, and payment history.

Decision Tree Learning -

A Decision Tree creates a hierarchy of decision rules. It selects important features and divides customers into groups. For example, the tree may first examine contract type, then monthly charges, followed by the number of complaints. The final leaf node provides the churn prediction.

The resulting rules can be easily examined by managers, making the model highly interpretable.

KNN -

KNN stores existing customer records. For a new customer, it calculates the similarity between the new customer and existing customers. It then selects the K most similar customers and uses their churn outcomes to make the prediction.

For example, if most of the nearest customers have left the service, KNN may classify the new customer as likely to churn.

Factor	**Decision Tree**	**KNN**
Interpretability	Very high; decisions can be represented as rules	Lower; based on neighbouring records
Training time	Generally fast	Very low because little model training is required
Prediction time	Usually fast	Can be slower for large datasets
Noisy data	Can overfit without pruning	Can be affected by noisy neighbours
Large datasets	Generally scales better	Prediction becomes computationally expensive
Complex feature spaces	Can handle non-linear relationships	Performance depends strongly on distance
Feature scaling	Usually not required	Usually important
Decision process	Rule-based	Similarity-based

Analysis -

Decision Trees are easier to interpret because the prediction can be explained through a sequence of rules. This is useful in customer churn analysis because a company can identify the factors responsible for churn.

KNN is simpler in terms of training because it does not need to build a complex model. However, every new customer may need to be compared with many stored customers, making prediction slower as the dataset grows.

Decision Trees can also handle a mixture of important and less important features more effectively, while KNN can be strongly affected by irrelevant features and differences in feature scales. Both methods can model non-linear classification patterns, but they approach them differently.

Conclusion: Decision Trees are generally more suitable for large-scale customer churn prediction when interpretability, fast prediction, and rule-based decision-making are important. KNN is useful when similar historical customers provide strong predictive information and the dataset is manageable in size.